A PROJECT TITLE

ON

Heart Disease Prediction Using Logistic Regression, SVM, and K-Nearest Neighbors

Project report submitted in partial fulfilment of the requirements of the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

In

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By

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1.Abstract:

Heart disease is one of the major health concerns worldwide, and early detection can play an important role in reducing the risk of severe cardiovascular complications. Traditional diagnosis generally depends on medical examinations, clinical measurements, and the experience of healthcare professionals. With the growth of machine learning, healthcare datasets can be analyzed to identify patterns and predict the possibility of heart disease more efficiently. This project presents a machine learning-based system for predicting the presence of heart disease using three classification algorithms: Logistic Regression, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN).

The proposed system uses relevant patient health parameters such as age, sex, chest pain type, resting blood pressure, cholesterol level, fasting blood sugar, maximum heart rate, exercise-induced angina, and other clinical attributes available in the selected dataset. The collected dataset is preprocessed by handling missing values, encoding categorical attributes, scaling numerical features where required, and dividing the data into training and testing sets. The three machine learning algorithms are then trained using the processed dataset to classify patients into appropriate heart disease categories.

The performance of Logistic Regression, SVM, and KNN is evaluated using standard classification metrics such as accuracy, precision, recall, F1-score, and confusion matrix. Comparing these algorithms helps determine which model provides better predictive performance for the given dataset. The system aims to demonstrate how machine learning techniques can support early risk prediction and provide useful insights from medical data. The proposed approach is intended as a predictive and educational tool and should not replace professional medical diagnosis. Overall, this project demonstrates the application of supervised machine learning techniques in healthcare and highlights the potential of data-driven methods for heart disease prediction.

2. INTRODUCTION:

Heart disease is a major health concern worldwide, and early prediction can help in taking preventive measures and providing timely medical attention. Machine learning can analyze patient health data and identify patterns related to heart disease.

This project uses three machine learning algorithms: Logistic Regression, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN) to predict the presence of heart disease. The models are trained and tested using a heart disease dataset and evaluated using metrics such as accuracy, precision, recall, F1-score, and confusion matrix.

The main objective is to compare these algorithms and identify an effective model for heart disease prediction. The system is intended as a supportive prediction tool and not as a replacement for professional medical diagnosis.

3. SOFTWARE REQUIREMENTS:

Operating System: Windows 10/11, Linux, or macOS

Programming Language: Python 3.x

Development Environment: Jupyter Notebook, Google Colab, or VS Code

Libraries: NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn

Algorithms: Logistic Regression, SVM, and KNN

Dataset: Heart disease dataset containing patient health and clinical attributes

Database: Optional – MySQL/SQLite for storing prediction records

Hardware: Minimum 4 GB RAM, Intel Core i3 or equivalent, and 5 GB storage

Internet: Required for Google Colab and package installation

Signature of the Guide

Course Instructor